Industrial Safety Processes

Hot & Cold Metal Working

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Metal Working Machines

- Presses / Forming machines
- Punches
- Drill Press
- Saws / Grinding
- Shears
- Metal Lathes
- CNC / Milling machines
- Water Jet & Laser Cutting

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Press / Forming Machines

- Form metal into various shapes using high pressure
- Can be manual – small jobs, thin / pliable stock
- High pressure hazard and real danger of losing body parts if in the area of operation

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Punches

- Usually limited to punching out blanks for other work
- Commonly running machines that operate a set rate with operators feeding stock
- Increased use of automatic machines similar to CNC / milling machines

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Drill Press

- Special bits and alloys used to cut various metals
- Sharpness and condition of bits critical to safety
- Usually don’t rotate as fast as woodworking drills
- Both manual and CNC / automatic systems
- Often require water or water / grit mixture to cool operation and aid in cutting

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Metal Saws

- See previous portion of the course on grinders
- Dedicated metal saws usually have larger diameter cutting disks and are similar to radial arm saws
- Metal band saws – fixed and portable – are also common

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Shears

- Major shear point hazard
- Guarding critical as operator access usually required at some point
- Larger feed tables can assist in keeping operator distance
- Ideally systems should be automated as much as possible

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Metal Lathes

- Concern with flying metal and potential for operators being caught in rotating systems
- More commonly seen in CNC configurations – especially in smaller shops
- Controlling chips and cooling shaping area are important

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CNC / Milling Machines

- More common – even smaller industrial operations
- Permits specialized metal formation to very high standards
- Milling often understood as “open” operation that is not contained inside the machine – but that is changing

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CNC Machining

- Computer controlled metal fabrication systems
- Standalone machinery that is designed to totally enclose and separate operations from workers
- Combination of lathe, shaping, drilling, milling, cutting, etc. options – increasingly with single machine
- Interlocks on access doors represent a major shift in safety in metal operations

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Water Jet Systems

- Newer method of metal cutting
- Essentially uses a jet of water with an abrasive mixture to cut through metal and other hard materials
- Alternative to punches and other metal forming due to programming flexibility
- Very precise cutting of metals and other materials

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Laser Cutting

- Replacing water jets in several applications
- Considered more precise that water cutting
- High efficiency without the need for application of granular mixture for water jets
- Good primarily for metal – not other materials (plastics)

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Machine Design

- Design of machinery and operation is an important factor in determining operator safety
- Some designs and operation are inherently more dangerous and likely to result in injury
- Machinery operation and applied safeguards need to be considered together

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Safety Based on Operation

- How a machine functions will help determine best safeguarding approaches
- Nature and extent of worker access is a major consideration
- Nature and extent of material entry/exit is a concern

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Brake

- System on the press that stops or holds the operating mechanism on the press either directly or by disengaging the clutch

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Full-Revolution Clutch

- A type of press clutch that, when tripped, cannot be disengaged until the drive mechanism has completed a full revolution, operation, rotation, or stroke of the mechanism
- Will not stop if actuator is released during operation

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Part-Revolution Clutch

- A type of clutch that can be disengaged at any point before the drive mechanism has completed a full revolution, cycle, rotation, or stroke of the operating mechanism

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Anti-Repeat Operation

- The part of the clutch/brake control system designed to limit the press to a single stroke if the actuating means is held or stuck to the “on” or “operate” position
- Requires release of the actuating means before another stroke can be initiated
- Also known as “single-stroke reset” or “reset circuit”

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Safety In Operation

- Part-revolution clutch systems are safer to use than full-revolution clutch systems
- Anti-repeat systems can be critical for worker safety
- Operational safety can be enhanced by using appropriate safeguards that protect based on the operation of the unit (anti-repeat or not, full or part-revolution clutch)

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Types of Physical Guarding

- Fixed die-enclosure guards
- Fixed barrier guards
- Interlocked press-barrier guards
- Adjustable barrier guards

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Type-A Moveable Barrier Devices

- Type-A Moveable Barrier Devices protect the operator by enclosing the point of operation before a press stroke begins
- The barrier stays in place until the motion of the machine has stopped at the end of its cycle
- Should be used for full-revolution machines or machines without anti-repeat

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Type-B Moveable Barrier Device

- Type-B Moveable Barrier Devices protect the point of operation before a press stroke begins
- This device can be opened on the up-stroke or return motion of the machine, or if down-stroke or action motion is interrupted
- Useful for quick movement of material and stock
- Dangerous and not recommended to be used with full-revolution machines

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Other Types of Safeguarding

- Two-hand operation
- Tie-backs
- Pull-backs
- Electronic sensing devices
- Hand/foot controls

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Foundries – Main Activities

- Heat metals to melting point
- Mix liquid metals
- Pour liquid metals into moulds to create products (casting)

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Heat

- Extreme temperatures for personnel and equipment
- Equipment needs to be well maintained and used as specified
- Specialized PPE can be critical – heat, noise, impact
- Some vessels require pre-heating to avoid damage
- Potential for steam explosions from water sources

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Hoists and Cranes

- From large systems to manual monorails
- Visibility a key concern due to lighting difficulties
- Movement of molten metal and heated metal pieces
- Limited space means extra caution needs to be taken to avoid interactions between loads and personnel

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Material Handling

- Manipulation of ladles and crucibles
- Scraping slag from vessels
- Often manual or only partially mechanical movement of products and finished items
- Exacerbated by physical impact of heat on human body

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Lighting

- Very difficult to properly light foundry operations
- Contrast in light from molten metal against background
- Large spaces
- Smoke, dusts, sand coating light fixtures

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Fire Protection and Control

- Accumulation of dusts that may be combustible or explosive
- Separation of high-heat areas and incompatible operations (slag, water, different heating systems)
- Ensuring ovens, furnaces, cupolas are well-maintained and do not fail

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Noise

- Furnace, oven, cupola firing systems
- Air blowers and O_2 injection systems
- Cranes, lifts, PITs
- Steam, metal flow
- Forming equipment – sand mills, forges / hammers, casting compactors

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Housekeeping

- Visibility can be impaired by visual contrasts, poor added lighting, smoke / steam, PPE
- Cleaning up prevents accumulation of dusts (vacuuming preferred) and debris
- Limits potential for incidents and reduces issues with product quality

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Ventilation

- Helps with heat control and airborne contaminants
- Really limited in terms of dust extraction due to nature of work and limitations of systems – housekeeping program critical
- Concern with accumulation of dusts in ventilation posing fire / explosion hazard – particularly aluminum

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Casting

- Increasingly involves specialized machinery and materials
- Not uncommon for addition of chemicals and oils as part of casting process
- Can often involve close proximity to molten metal
- Machinery can pose additional ergonomic, noise, and guarding issues

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Health Hazards

- Heat stress
- Over exertion
- Dusts – respiratory
- Dermatitis can develop with longer exposures to solvents, oils, dusts, acids, etc.
- Exposure to metal fumes

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Addressing Health Hazards

- Ventilation controls
- Work rotation and limits
- Personal protective equipment
- Monitoring systems
  - Environmental
  - Biological

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